

Erik Earl

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Experience

Software Engineer, Platform & DevOps | Qualcomm

April 2025 – Present

- Designed and built a Kubernetes native Database as a Service platform, replacing a manual VM provisioning process with an API driven, self service workflow for on demand database deployment and lifecycle management
- Built a custom Kubernetes operator (Python, Kopf) deployed across a multi cluster environment, implementing the control loop reconciliation pattern for 3 CRDs (MySQL, PostgreSQL, MongoDB) to orchestrate cross cluster database provisioning, credential lifecycle, and teardown via REST API calls to backing database operators
- Implemented cross cluster backup and restore workflows, enabling teams to provision databases from backups across environments through automated S3 backup discovery and credential provisioning
- Supported Kubernetes networking through an RKE1 to RKE2 migration, including a CNI transition (Calico to Cilium), managing ingress and Gateway API routing, TLS/certificate management, and firewall/VIP traffic flow
- Developed a custom ExternalDNS webhook provider for BlueCat DNS, enabling automated DNS lifecycle management for Kubernetes resources while reducing API load through a write-through caching layer
- Administered a shared Jenkins CI/CD platform, supporting teams by developing reusable pipeline libraries, onboarding new teams, and providing operational support for build, deployment, and infrastructure incidents

Software Engineer Intern | Qualcomm

May 2024 – Aug 2024

- Built an automated TLS certificate management system (Kubernetes, Cert-Manager), automating certificate issuance, renewal, and rotation, reducing manual overhead and improving security compliance
- Developed a Python Kubernetes operator to manage certificate lifecycle and automate resource provisioning
- Designed and implemented Jenkins CI/CD pipelines for automated testing and deployment
- Worked in Linux based environments using CLI tools to manage containerized, cloud native systems

Software Engineer Intern | Qualcomm

May 2023 – Aug 2023

- Implemented a backend IP lifecycle management system, reducing manual approval processes by ~80%
- Developed REST APIs using Python and Django to support a distributed, multi-service architecture

Software Engineering Co-op | Lubrizol Corporation

May 2022 – April 2023

- Developed internal automation tools and software solutions using Bash, C++, JavaScript, and web technologies

Education

B.S. Computer Science, Minor in Artificial Intelligence | *Cum Laude*
Case Western Reserve University | *Cleveland, OH*

Dec 2024
GPA: 3.85 / 4.00

Projects

Containerized Minecraft Server Deployment Platform (Home Lab)

- Designed and implemented a Kubernetes native platform to provision and manage containerized Minecraft servers via custom resources, automatically creating Deployments, Services, and PersistentVolumeClaims
- Developed a Kubernetes operator and CRDs to automate server provisioning, enabling fast deployment of servers at scale through a configurable, declarative interface mapped to container runtime settings
- Built a Flask based control plane exposing REST APIs to interface with the Kubernetes API, enabling server lifecycle management and routing requests between external clients and in-cluster services

Manga Translation with Machine Learning and Computer Vision (Senior Capstone)

- Developed a computer vision pipeline to automate manga translation with image detection & segmentation, optical character recognition (OCR), and language translation models
- Trained a binary image segmentation model in PyTorch using transfer learning with ResNet34 U-Net architecture

AI Data Platform & MCP Service for Crowd-Sourced Climbing Data (Home Lab)

- Developed a TypeScript based Model Context Protocol (MCP) server exposing AI accessible APIs for querying and analyzing over a semi-structured dataset of millions of crowd sourced climbing records from Mountain Project
- Built a containerized data collection platform on a self hosted Kubernetes cluster, leveraging CronJobs, shared NFS backed persistent storage, and automated data scraping and processing pipelines to continuously ingest, normalize, and serve datasets of records to the MCP service

Skills

Python · Go · TypeScript · SQL · Bash · Kubernetes · Docker · Containerization · Helm · Fleet · Terraform · Grafana
Prometheus · VMware · Networking · Cilium CNI · DNS · AWS (EKS, S3) · Jenkins · GitOps · Linux/Unix · Node.js

Certification: Certified Kubernetes Application Developer (CKAD) – The Linux Foundation